

Long type questions

1. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consist of (i) exactly 3 girls, (ii) at least 3 girls, (iii) at most 3 girls
2. In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other?
3. In how many ways can 7 people be seated such that two specific people are never adjacent?
4. How many integers between 1 and 500 (inclusive) are
 - a. Divisible be at least one of 3, 5, 7?
 - b. Divisible by 3 and 5 but not 7?
5. How many integer solutions exist for $x_1 + x_2 + x_3 = 10$ where $x_1 \geq 2$ and $x_2, x_3 \geq 0$?
6. Solve the recurrence relation $a_n = 3a_{n-1} + 2$, $a_0 = 1$, using any method.
7. Find the solution of recurrence relation $a_n - 3a_{n-1} = 2(3^n)$.
8. Use generating function to solve: $a_n = 2a_{n-1} + 1$, $a_0 = 0$
9. Solve the recurrence: $a_n = 3a_{n-1} - 2a_{n-2}$, $a_0 = 2$, $a_1 = 5$
10. Use generating function to solve the recurrence relation $a_r - 3a_{r-1} + 2a_{r-2} = 0$ for $r=2,3,4\dots$ and initial condition $a_0 = 2$, and $a_1 = 3$, What is the solution of the recurrence relation $a_n = a_{n-1} + a_{n-2} + 3n + 1$ for $n \geq 2$, with $a_0 = 2$ and $a_1 = 3$?
11. Solve the recurrence relation $a_n - 2a_{n-1} - 3a_{n-2} = 0$, $a_0 = 3$, $a_1 = 1$ by using generating function.
12. Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$, for all $n \geq 2$ by using generating function.
13. Find the sequence b_r from the generating function $\sum_{r=0}^{\infty} b_r x^r = \frac{1}{x^2 - 5x + 6}$

Short type questions

- a) Use the Pigeonhole Principle to prove: In any group of 13 people, at least two were born in the same month.
- b) How many permutations can be formed from the letters of the word "LEVEL"?
- c) What is the maximum number of groups into which you can distribute 30 students such that at least one group must have at least 5 students?
- d) Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$.
- e) How many ways can 8 people be arranged in a line if two specific people must stand together?
- f) How many ways can 5 people be seated in a circle?
- g) What is the pigeonhole principle? Give an example.
- h) Solve the recurrence: $a_n = 2a_{n-1}$, $a_0 = 1$.

- i) What is the maximum number of groups into which you can distribute 30 students such that at least one group must have at least 5 students?
- j) Calculate the number of 3-combinations from a set of 5 distinct elements.
- k) How many ways can 8 people be arranged in a line if two specific people must stand together?